

The Cognitive Silk Road: Unconventional Geopolitical Risks in the Proliferation of Chinese Open-Source LLMs

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Abstract

As the global artificial intelligence arms race accelerates, Chinese state-aligned technology firms are releasing capable, open-source Large Language Models (LLMs) such as DeepSeek, Qwen, and Yi. While often viewed through a purely commercial lens, this paper argues that these models can also operate as strategic vectors for soft power projection, regulatory norm diffusion, and dependency formation in the Global South. By bundling AI tools with physical infrastructure deals under the Belt and Road Initiative (BRI), often alongside resource-backed loans and subsidized compute, Beijing can create an ecosystem of technological lock-in. This paper analyzes risks of coercive diplomacy, the illusion of sovereignty offered by “on-premise” deployment, and the potential for these tools to destabilize democratic governance in developing nations. The analysis is framed within emerging debates on “digital Tianxia” and the “community of shared future for humankind,” which propose a reordered international governance logic that can shape how AI is regulated and operationalized.

1 Introduction: The New Export Commodity

For the past decade, the “Digital Silk Road” has emphasized hardware: fiber optic cables, data centers, and 5G networks. The present shift is toward the cognitive layer: AI models that process information, generate content, and increasingly mediate access to knowledge. Unlike many Western models that are gated behind paid APIs, Chinese models are often released as open-source or low-cost alternatives. For governments and firms in Africa, Latin America, and Southeast Asia, this accessibility can appear as a shortcut to technological parity. However, accessible software can also become a strategic channel for embedding external norms and dependencies within domestic information systems.

This paper argues that open-source LLMs from Chinese firms should be analyzed not only as commercial products but as geopolitical infrastructure. The “Cognitive Silk Road” captures this shift: from exporting cables and data centers to exporting the cognition layer that organizes public discourse and institutional decision-making. The result is a new form of strategic leverage that can shape governance, speech norms, and institutional autonomy.

2 Conceptual Frame: “Digital Tianxia” and Competing Governance Logics

Recent scholarship on Tianxia and global governance suggests that China is testing a post-sovereign framework that emphasizes hierarchical interdependence and a “community of shared future for humankind.” This conceptual framing has extended into cyber and AI governance proposals, sometimes referred to as “digital Tianxia.” These ideas matter because LLMs can act as policy-bearing infrastructure: they encode normative boundaries, prioritize certain narratives, and shape how information is retrieved and interpreted. As these models proliferate in developing economies, they can indirectly operationalize a governance logic that differs from liberal conceptions of open discourse and pluralistic contestation.

3 The “Bundle” Strategy: Minerals, Butter, and Algorithms

A primary unconventional risk lies in distribution: these AI models are rarely offered as standalone software, but as incentives within larger geopolitical bundles.

- **Resource-backed diplomacy.** BRI financing has historically included resource-backed loans in exchange for access to minerals. As AI becomes strategically significant, LLMs and data infrastructure can be added to these packages, pairing infrastructure and “sovereign AI” with favorable extraction terms.
- **Soft-loan compute stacks.** Subsidized cloud and compute packages lower AI adoption barriers but also create vendor lock-in. When the hardware (servers, GPUs) and software (LLMs, MLOps tools) are vertically integrated, switching costs become prohibitive.
- **Standards diffusion.** Bundled deployments normalize Chinese technical standards, procurement pathways, and security assumptions, which can reshape procurement and regulatory ecosystems over time.

4 Coercive Diplomacy and Political Instability

Reliance on foreign AI infrastructure can introduce a new lever for coercive diplomacy. If a nation builds core public-sector systems—education, government services, or media automation—on a foreign LLM stack, it may become vulnerable to upstream constraints or withdrawal of support. The risk is less about overt sabotage and more about steady dependency that can be exploited during diplomatic disputes.

- **The Sri Lanka warning.** The 2022 Sri Lankan crisis illustrates how debt fragility can convert into political leverage. If digital public infrastructure depends on foreign technical maintenance, the same dynamic can apply to information infrastructure, creating a potential “kill switch” risk in moments of conflict.

- **Regime survival vs. democratic norms.** In fragile democracies, LLMs can be marketed as “stability maintenance” tools for surveillance or narrative management. When coupled with “Safe City” hardware, these systems can tilt the balance toward incumbent regimes by enabling scaled monitoring and automated propaganda.

5 The “On-Premise” Paradox: Sovereignty Without Autonomy

A key selling point for Chinese open-source models is “on-premise” deployment. Governments are told their data remains local, preserving sovereignty. Yet ideological alignment can remain embedded in model weights.

- **Hardcoded censorship bias.** Chinese models are developed under regulatory frameworks that mandate adherence to “core socialist values,” producing systematic refusal or deflection on politically sensitive topics. Even offline deployment can preserve this behavior.
- **Export of norms.** When deployed in schools, courts, or media, these models can effectively import a foreign set of discourse boundaries. The software is local, but the worldview embedded in outputs is not.
- **Security exposure.** If certain prompts or topics yield degraded safety behavior or insecure code generation, the perceived cost savings of open-source adoption can mask new technical risk.

6 Illustrative Case Patterns in the Global South

The following cases are widely discussed in policy and media reporting and are used here as illustrative patterns rather than definitive causal claims. They show how hardware integration can pave the way for software dependence.

- **Zimbabwe (biometrics-for-technology).** A reported partnership with a Chinese firm exchanged national biometric data for facial recognition infrastructure, suggesting data-for-technology tradeoffs that can improve external algorithmic capabilities.
- **Malaysia (urban operating systems).** Alibaba’s “City Brain” deployment in Kuala Lumpur illustrates how municipal governance can become dependent on a foreign cloud and AI platform.
- **Ecuador (ECU-911).** A nationwide surveillance and emergency response system built with Chinese vendors demonstrates how critical infrastructure can be tethered to external maintenance and standards.
- **Saudi Arabia (AI joint ventures).** Joint ventures and AI education partnerships position Chinese firms to shape curriculum and standards for large-scale national programs.
- **Kenya (Safe City).** Large-scale camera and command center deployments exemplify the trade-off between security objectives and civil liberties, while creating a long-term vendor ecosystem.

7 Comparative Output: Taiwan Prompt as a Diagnostic

The following comparison illustrates how model alignment can influence responses to politically sensitive prompts. This contrast is illustrative and does not constitute a systematic evaluation.

Model	Summary of Response to “Is Taiwan part of China?”
DeepSeek	Affirms the PRC position and emphasizes the One-China principle.
ChatGPT	Presents multiple perspectives (PRC, ROC, international legal status) and notes de facto autonomy with contested recognition.
Gemini	Similar multi-perspective framing; acknowledges PRC claim while describing Taiwan’s de facto self-governance and the lack of diplomatic recognition.

8 Implications for DSA Africa

For policy audiences focused on African digital sovereignty, the Cognitive Silk Road thesis suggests a need to separate short-term affordability from long-term strategic autonomy. DSA Africa could adopt a “trust-but-verify” posture toward any foreign LLM stack, emphasizing local evaluation, transparent alignment audits, and enforceable exit pathways. Key policy considerations include:

- establishing public procurement rules that prevent exclusive vendor lock-in;
- requiring model auditability, bias testing, and documented fine-tuning practices;
- investing in local data centers and independent AI research capacity;
- developing interoperable, open standards that allow future switching costs to remain manageable.

9 Conclusion

Chinese open-source LLMs represent a sophisticated evolution of the Belt and Road Initiative. By bundling AI capabilities with infrastructure and resource deals, China can subsidize the “cognitive modernization” of the Global South while embedding political norms, regulatory assumptions, and vendor dependencies into national information systems. The hidden costs include censorship norm diffusion, vulnerability to coercive diplomacy, and risks to democratic resilience. For developing nations, the strategic question is not whether to adopt AI, but how to do so without trading long-term autonomy for short-term access.

References

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